DAY 40 · THEORY HANDOUT

Ask a Small Database in Plain English

Follow schema, SQL, validation, execution, and result steps.

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The idea and its useful terms

Follow schema, SQL, validation, execution, and result steps.

SQL

A language for asking a database for rows.

Schema

The database table names and column definitions.

Read-only connection

Lets our class query data without changing rows.

The model needs the schema to use real table and column names. It can still generate wrong SQL, so we check before execution.

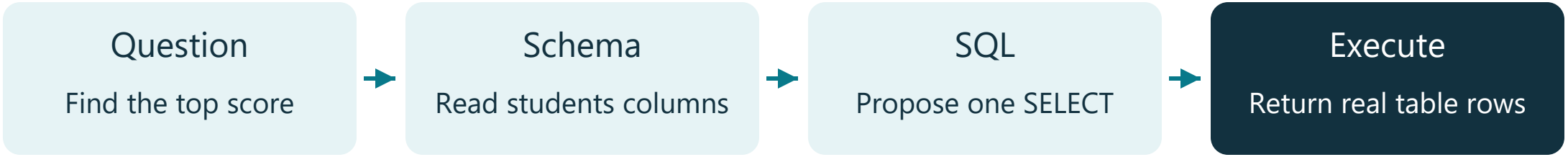
This lesson uses a tiny local SQLite database, as the course example does. MySQL uses a server and credentials, but the main question-to-query steps are the same.

> A prompt saying "only SELECT" is not a security boundary. Database permissions must enforce read-only access.

Follow the work, one step at a time

Each step has a job. We can inspect its output before moving on.

Read the path from left to right.



Each arrow means: pass the result to the next step.

The schema lists students with name and score columns. The model proposes SELECT SQL using those names. Validation checks the simple allowed form. The read-only connection runs it and returns real rows.

The source course also separates query repair after an error. A repair must be checked again. Our database authorizer restricts tables and operations even if model instructions are ignored. The example has only three rows.

> If the answer is wrong, inspect the input and the intermediate results. The last sentence alone cannot show which step failed.

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A small example we can check

The values here are classroom examples. The notebook shows actual results.

Table

students(name, score)

Asha 82, Ravi 70, Mia 91.

Question

Who has the top score?

Use the score column.

Checked SELECT

ORDER BY score DESC

LIMIT 1 returns Mia 91.

Read the given input first. Point to the fact or value needed for the requested output.

Run the matching example, then try the fresh input at the end of the notebook. Compare both results with the known input.

> A prompt saying "only SELECT" is not a security boundary. Database permissions must enforce read-only access.

Choose the setting or tool for its job

A different choice changes the work. These are the choices we compare.

Inspect schema

Find names and types.

Do this before guessing SQL.

Validate and restrict

One SELECT, known tables.

Read-only DB is the boundary.

Repair once

Use the error + schema.

Recheck before another run.

Inspect a three-row table, generate a SELECT query, and display it before running.

Use a read-only connection and an allowed-table check. Test a rejected UPDATE.

> Create an agent with schema and query tools. Try a different score question and inspect its tool result.

Practice, check, and read the full lesson

Use the notebook to try the steps. Use the article when you need the longer explanation.

1. Build

Run the small example.

Inspect each step.

2. Change

Try the fresh input.

Keep the goal clear.

3. Check

Compare with the source.

Keep mistakes visible.

Create an agent with schema and query tools. Try a different score question and inspect its tool result.

Read the complete tutorial breakdown for this section in our KGP Talkie article.

> A prompt saying "only SELECT" is not a security boundary. Database permissions must enforce read-only access.

DAY 40 · COMPLETE

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